

Patrick Rizzardi

Senior Backend Engineer & API Architect

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SUMMARY

Senior backend engineer who builds and scales production systems from scratch. Sole architect of a workforce platform that grew from zero to 100,000+ users and \$2M+/month in cashflow. I ship the integrations other teams call "too complex," and go deep on the layers most engineers only consume — distributed systems, compilers, and from-scratch ML. TypeScript/Node and MySQL/PostgreSQL primary; Rust when the problem demands it. 12 years military (NCO) — I own outcomes and don't need managing.

CORE SKILLS

Languages	TypeScript, Rust, Python, SQL, JavaScript, PHP
Backend	Node.js, Bun, Fastify, Express, REST APIs, Webhooks, OAuth, Microservices
Data	MySQL, PostgreSQL, Redis / Redis Streams, Sequelize
Infra	Docker, GitHub Actions, AWS, DigitalOcean, Linux, CI/CD
Frontend	Vue 3, React, Tailwind, Vite
Systems / ML	LLVM, Compilers, CUDA, Transformers

EXPERIENCE

Senior Software Engineer & Co-Lead — VPM Solutions

Feb 2021 – Present · Remote

- Sole **backend, database, and infrastructure engineer** — built the entire REST API and MySQL data layer from scratch and **re-architected the AWS infrastructure** into a redundant, fault-tolerant setup, scaling the platform from **zero to 100,000+ users and \$2M+/month in cashflow**.
- Grew the API to **88 data models and 282 controllers**; promoted to co-lead — now drive technical direction and mentor a team of 4 engineers.
- Built and run the **Stripe** payment infrastructure — payments, payouts, identity verification, **\$65.8M+ gross volume since 2021** (migrating contractor payouts to Wingspan for lower fees + faster global delivery); shipped 10+ integrations end-to-end overall (incl. Hubstaff), several solo — idempotent, retried, logged.
- Designed and built the production infrastructure — Docker orchestration, CI/CD, redundant fault-tolerant services — with **no unplanned outages in over a year**.

Founder & Principal Engineer — Redact Digital

Aug 2024 – Present · Remote

- Trading Platform v3** — fault-tolerant automated trading system: 6 microservices on Redis Streams with exactly-once delivery; a 300–500M-row partitioned database serving every query under 50ms (window functions turned a 12s query into 40ms).
- Yinz** — a compiled systems language targeting LLVM native code. Full compiler in Rust (lexer, parser, type checker, LLVM codegen via inkwell), incremental recompilation via salsa, plus an LSP server, formatter, watch daemon, and VSCode extension.
- Tessa AI** — a from-scratch LLM training stack in Rust implementing three architectures (LLaMA-style transformer, RWKV-7, hybrid), custom CUDA attention kernels (2–3x throughput), and a BPE tokenizer trained on a 34GB / 3.85B-token corpus.
- Error Decoder** — monetized developer product (browser extension + web app) that decodes production errors from React, Vue, Angular, and Node.js. **YinzerFlow** — published npm framework, security-first, 66+ tests.

Freelance Developer — Self-Employed

2018 – 2021 · Remote

- Delivered web and API projects for small businesses, owning each end-to-end — from scoping requirements with non-technical clients through deployment.
- West County Fitness** — full gym website with a Google Calendar API integration (owner self-service scheduling) and a custom blog CMS (Laravel/PHP + React), self-hosted on DigitalOcean.
- Shipped **5–6 custom WordPress sites** on ~1-week brief-to-production turnarounds.